Feiyang Wu

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Research Assistant, CUHK-Shenzhen | PCSELs · Computational Photonics · Research Software

Education

Shandong University, B.Sc. in Physics Sep. 2021–Jun. 2025
Taishan College honors physics track, national initiative for fundamental disciplines. GPA: 82.9/100.

Research Interests

Semiconductor lasers and micro/nano-optics; photonic crystal surface-emitting lasers (PCSELs); physics-based modeling, numerical validation, and automated inverse design.

Research Experience

PCSEL Device Research and Computational Modeling Aug. 2025–Present
Research Assistant, CUHK-Shenzhen; supervised by Prof. Zhaoyu Zhang

- **Full-wave design:** Build 3-D GaAs PCSEL models in Lumerical FDTD and COMSOL; analyze band structures and Γ-point modes; sweep lattice, hole, and epitaxial parameters.
- **Theory and verification:** Develop Python coupled-wave theory (CWT) eigensolvers; examine mode identity, spatial and Fourier convergence, and agreement with independent implementations; document model limitations and outstanding validation.
- **Automation and inverse design:** Connect solvers through Python/Java APIs and organize workflows from structural parameters through simulation and mode diagnostics to reports. Study value-based reinforcement learning and Bayesian optimization with Q-fit reliability gating, resulting in two co-first-author preprints.
- **Experiments and fabrication:** Build photoluminescence setups; participate in optical/electrical pumping and L–I–V, spectral, and beam measurements. Contribute to optical-pumping PCSEL fabrication, ICP recipe tuning, KLayout masks, and SEM-based process analysis.
- **Research coordination:** Assist with grant proposals, progress and closeout reports, and technical plans for device performance evaluation.

Mode Analysis and Phase Matching of LNOI/LTOI Waveguides Jan. 2024–Jun. 2025
Shandong University; supervised by Prof. Lei Wang

- Independently construct COMSOL FEM models to study how ridge-waveguide geometry affects effective indices and phase-matching wavelengths; investigate periodic poling for fundamental-mode phase matching.
- Participate in thin-film device grinding and polishing and the construction and operation of optical resonator frequency-response measurements.

Dynamic Memristor Reservoir Computing and Multimodal Recognition Oct. 2022–Oct. 2023
Shandong University; supervised by Prof. Limei Zheng

- Process text and speech recognition datasets, implement neural-network training workflows in MATLAB, and debug code for multimodal recognition and spatiotemporal learning studies.

Numerical Simulation of Black Hole Accretion Summer 2022
Shanghai Astronomical Observatory, CAS; supervised by Prof. Feng Yuan

- Study accreting gas using the ZEUS hydrodynamics code.

Preprints

1. L. Wen*, **F. Wu***, J. Yu*, C. Yuan, R. Li, Z. Zhang.
   When Every Simulation Counts: Value-Based Reinforcement Learning for Accelerated Photonics Inverse Design. arXiv, 2026. arXiv:2607.23469 (*Equal contribution)
2. J. Yu*, **F. Wu***, L. Wen*, C. Yuan, R. Li, Z. Zhang.
   Reliability-Aware Bayesian Optimization of 1310 nm PCSELs with FDTD Verification. arXiv, 2026. arXiv:2607.21772 (*Equal contribution)

Journal Publications

3. F. Nie, J. Wang, H. Fang, S. Ma, **F. Wu**, et al.
   Ultrathin $SrTiO_3$-based oxide memristor with both drift and diffusive dynamics as versatile synaptic emulators for neuromorphic computing. Materials Futures, 2023, 2(3): 035302.
   DOI: 10.1088/2752-5724/ace3dc
4. F. Nie, H. Fang, J. Wang, L. Zhao, C. Jia, S. Ma, **F. Wu**, et al.
   An Adaptive Solid-State Synapse with Bi-Directional Relaxation for Multimodal Recognition and Spatio-Temporal Learning. Advanced Materials, 2025, 37(17): e2412006.
   DOI: 10.1002/adma.202412006

Patent application: Z. Zhang, **F. Wu**, S. Mao, K. He. *Semiconductor Laser Automated Design Optimization Method and Storage Medium.* CN 202610820592.4; Jun. 8, 2026; CUHK-Shenzhen. Application accepted; not granted.

Research Software and Knowledge Resources

- **PCSEL computational tools:** Develop CWT solvers, COMSOL–finite-CWT batch workflows, and research-agent orchestration; standardize parameters and units and retain mode diagnostics, native-result indices, and validation status for traceable calculations.
- **DataSystem:** Internal Node.js/Python/SQLite tools for experimental, fabrication, and test records, with incremental imports, batch comparisons, and traceable reports.
- **PCSELBook and teaching tools:** Compile AI-assisted technical notes on Maxwell/Bloch analysis, CWT, PWEM, RCWA, FDTD, FEM, and carrier, thermal, and electro-optical coupling. Knowledge site; Optical Lab provides interactive geometric and paraxial optics demonstrations

Technical Skills

- **Modeling:** COMSOL, Lumerical FDTD; CWT, FDTD, FEM, PWEM, RCWA; band structures, cavity modes, parameter sweeps, and convergence verification.
- **Programming:** Python, MATLAB, Java APIs; research orchestration, data processing, automated reports; LaTeX, Mathematica.
- **Laboratory:** Photoluminescence, optical/electrical pumping, L–I–V and spectral/beam characterization; ICP etching, KLayout, SEM analysis.

Awards

- National Second Prize, China Undergraduate Physics Experiment Competition — Jul. 2023
- National / East China Regional Second Prizes, China Undergraduate Physics Tournament — 2022
- First Prize, Special Talent Scholarship (Research Innovation), Shandong University — Oct. 2023
- Third Prize, Academic Scholarship, Shandong University — Oct. 2024

Outreach and Leadership

- Share physics notes on Zhihu (since Sep. 2021); contribute electricity and nonlinear optics sessions to the Taishan Seminar (since Mar. 2022).
- Co-found the Physics Innovation and Competition Alliance at Shandong University (Aug. 2024), supporting competition training, academic exchange, and outreach.

Languages

Chinese (native); English: IELTS 6.5 (Listening 6.0, Reading 6.5, Writing 6.0, Speaking 6.5).